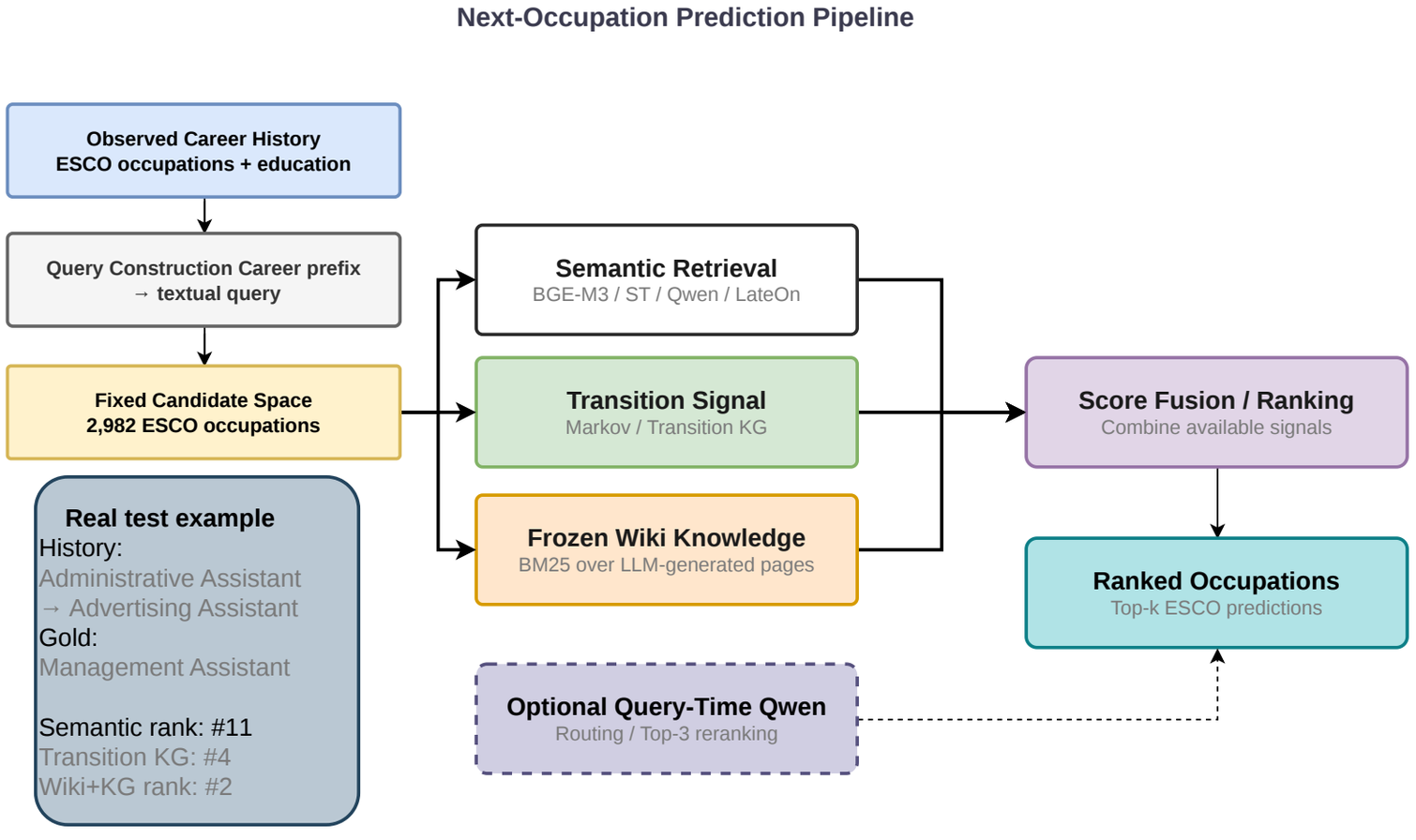
Next-Occupation Prediction Pipeline
Observed Career History
ESCO occupations + education
Query Construction Career prefix
→ textual query
Fixed Candidate Space
2,982 ESCO occupations
Semantic Retrieval
BGE-M3 / ST / Qwen / LateOn
Transition Signal
Markov / Transition KG
Frozen Wiki Knowledge
BM25 over LLM-generated pages
Score Fusion / Ranking
Combine available signals
Ranked Occupations
Top-k ESCO predictions
Optional Query-Time Qwen
Routing / Top-3 reranking
Real test example
History:
Administrative Assistant
→ Advertising Assistant
Gold:
Management Assistant
Semantic rank: #11
Transition KG: #4
Wiki+KG rank: #2